

B2B I

: v1.0 | : 2026-07-28 | : &

1.

B2B
(Incremental Cash Margin)' .1 , . . . ,

2. (Trade-off Matrix)

2.1 (Frontend)

&		(Trade-off)	
React 19 + Vite	B2B HMR	SPA ,	vs Next.js (SSR) B2B SEO SSR SPA
TanStack Query v5	API ,	, ,	vs Custom Fetch / Redux Thunk UI
URL Search Params	(2F~1F), URL	, D-Day	vs In-memory state , URL 100%
Zustand	60fps Multi-Select ,	, ,	vs Context API / Redux Context API Selector
TanStack Table v8	SKU , ,	, ,	vs Basic Table / AG Grid Headless (Virtualization)
Recharts	vs vs	, ,	vs Chart.js / D3.js React SVG
Tailwind CSS v4 + shadcn/ui	(#0F4C3A)	, ,	vs Styled Components / MUI CSS
React Hook Form + Zod			vs Formik / Plain State Zod
Web Vitals	LCP < 1.2s, INP < 100ms	UX	vs

Vitest + RTL + Playwright	, E2E	vs Jest / Cypress Vite 100% E2E
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2.2 (Backend) &	?	(Key Benefits)	&
Java 21 + Spring Boot 3.4	REST API ,	• Virtual Threads I/O • •	vs Node.js / Python SAP ERP/WMS
Spring Data JPA + QueryDSL 5.x	CRUD (, , D-Day)	• Q-Class • •	vs Native SQL / MyBatis SQL
Explicit Transition Service	➡➡➡ , @Version	• • • 1	vs Spring State Machine Enum + DB
Spring Batch 5 + Spring @Scheduled	02	• (Restart) • (Chunk) •	vs Quartz
Spring Security OAuth2 / OIDC	SSO/IdP , vs (RBAC)	• SSO • (RBAC) •	vs JWT IdP

2.3 (Database & Cache)	&	(Trade-off)
PostgreSQL 16+	+ JSONB	vs MySQL 8.0 / Oracle 19c 100% CBO JSONB GIN
Redis 7.x	, , LLM Caching (0)	vs In-memory HashMap TTL

2.4 AI (AI Service)	&	(Trade-off)
Python 3.11+ + FastAPI	ML	vs Spring AI / Spring Boot HTTP
PuLP / SciPy	· ·	vs
scikit-learn / LightGBM	ML	vs RAG / Vector DB ML 100% DB
Google Gemini 1.5 / Ollama /	, ,	vs LLM (Ollama)

GPT-4o-mini	API	0
	(Gemini	1,500)

2.5 , , Sentry (Observability & Testing)

&			(Trade-off)
Sentry	/	vs	(5,000)
Prometheus + Grafana	CPU, , DB RPS/Latency	vs CloudWatch	
Loki + Promtail	Logback JSON Grafana	vs ELK Stack	1/10
OpenTelemetry + Jaeger	React→Spring Boot→FastAPI→PostgreSQL	vs TraceID	TraceID
k6	500 Latency TPS	P95 vs JMeter / Locust	CLI

3. AI & 4

- ML (Feature Engineering) & : , D-Day, 30/60/90 , , LightGBM .
- / SKU (Cold-Start) : '→ ' , ' 65% ' .
- LLM (Structured Output) & Redis 0 : LLM JSON , Redis 0 .
- (Closed-Loop Feedback System): (,) DB AI (MAPE) , ML .

4.

Frontend (React SPA) Reverse Proxy (Nginx) Core Backend (Spring Boot) AI Microservice (Python FastAPI) Database (PostgreSQL/Redis) Observability (Prometheus/Loki/Sentry) .